

Ethanol Blending in Gasoline – United Kingdom

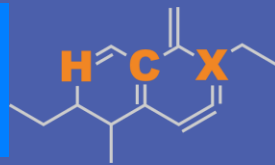
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**U.S. GRAINS &
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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in United Kingdom

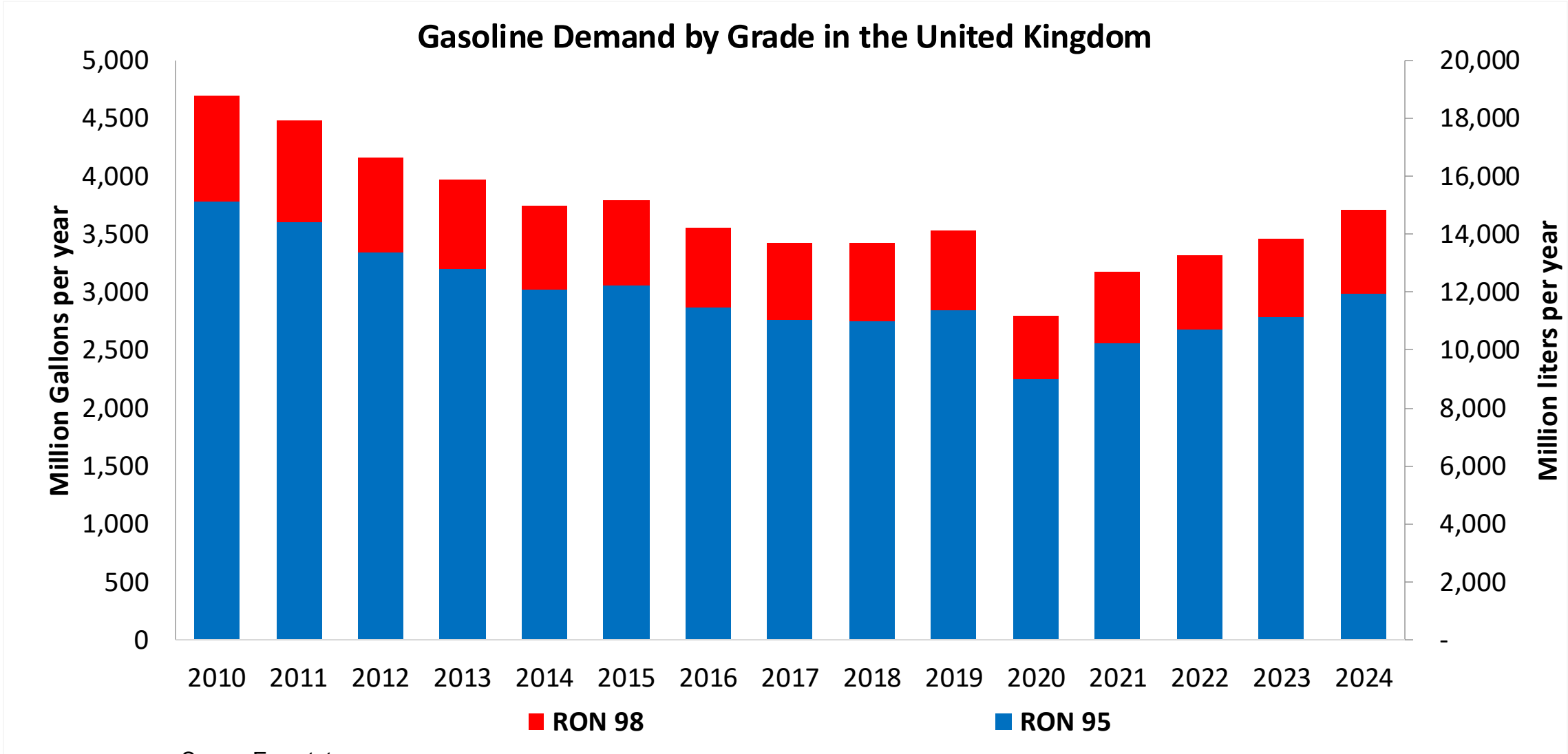


In 2024, gasoline consumption in United Kingdom was 14,052 million liters (3,712 million gallons) and growth rate was 1.0%. Gasoline production and net imports were 19,997 and -5,945 million liters (5,283 and -1,571 million gallons) correspondingly. United Kingdom complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.5 RON.

In 2024, ethanol consumption in United Kingdom was 1,105 million liters (292 million gallons) representing 7.9% of gasoline demand. Ethanol production and Net Imports were 177 and 929 million liters (292 and 245 million gallons) correspondingly. Ethanol sources are Corn 84%, Cereals 13% and Sugar Beet 3%.

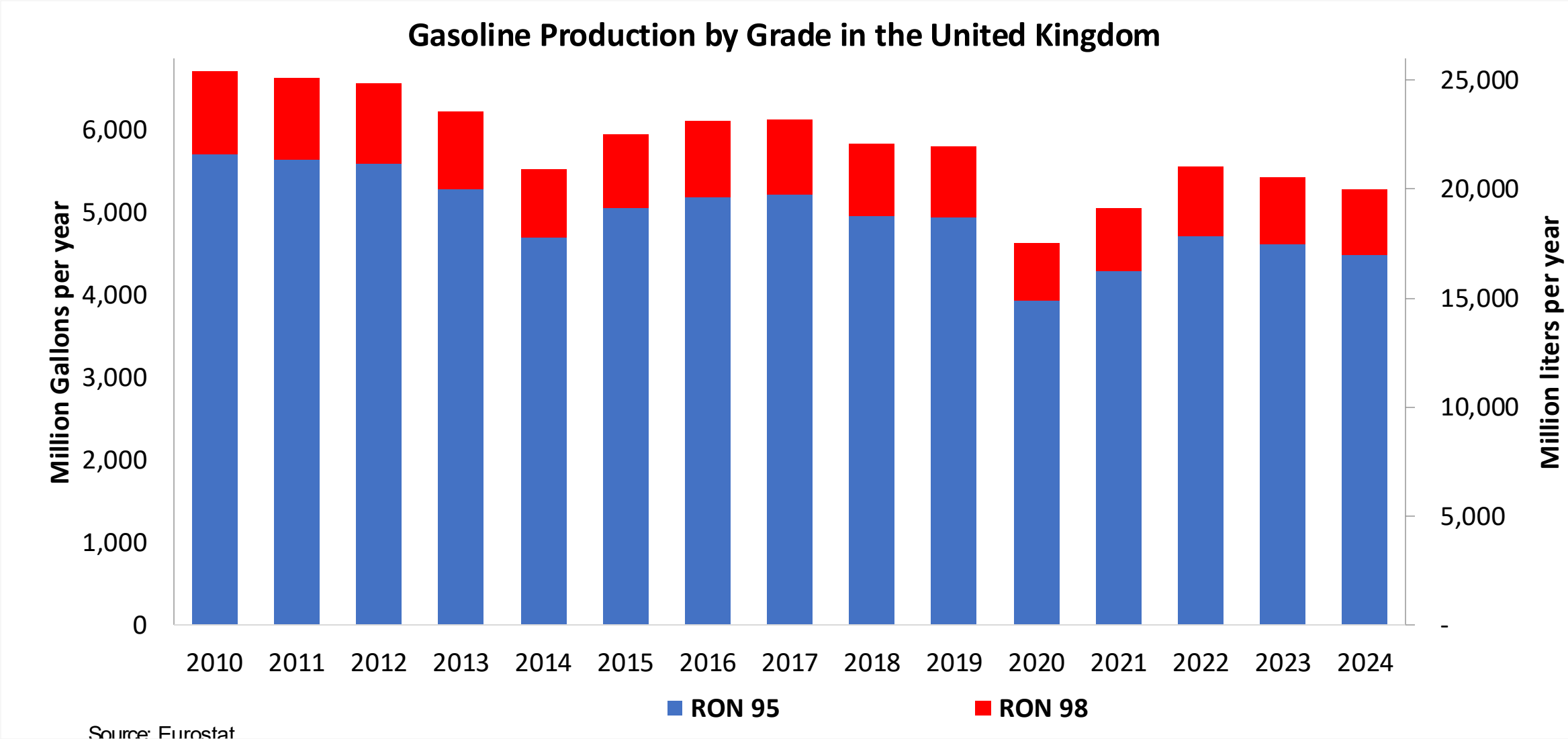
Source: Eurostat, European Commission

Gasoline Demand in United Kingdom

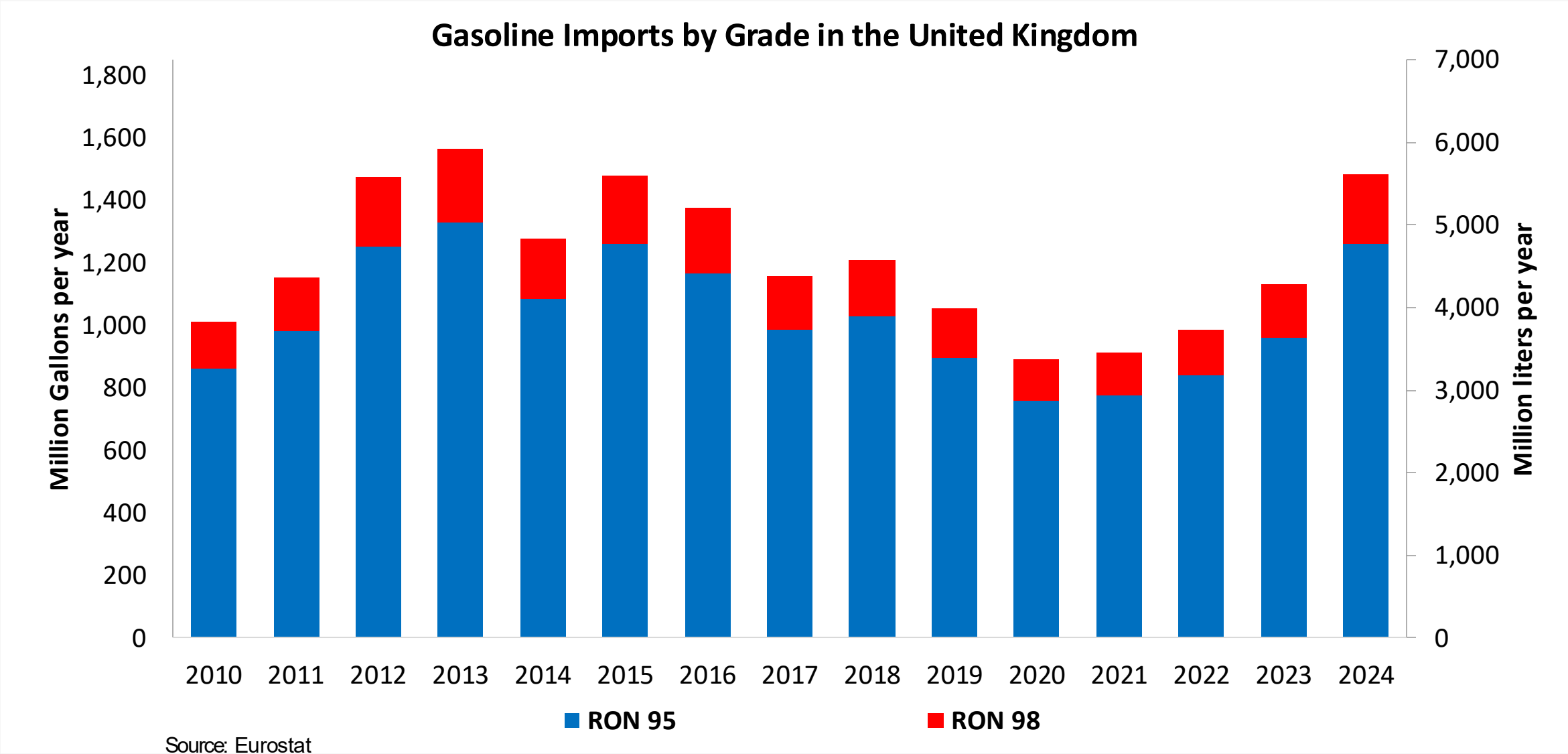


Source: Eurostat

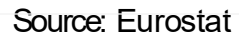
Gasoline Production in United Kingdom



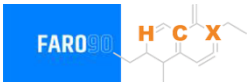
Gasoline Imports in United Kingdom



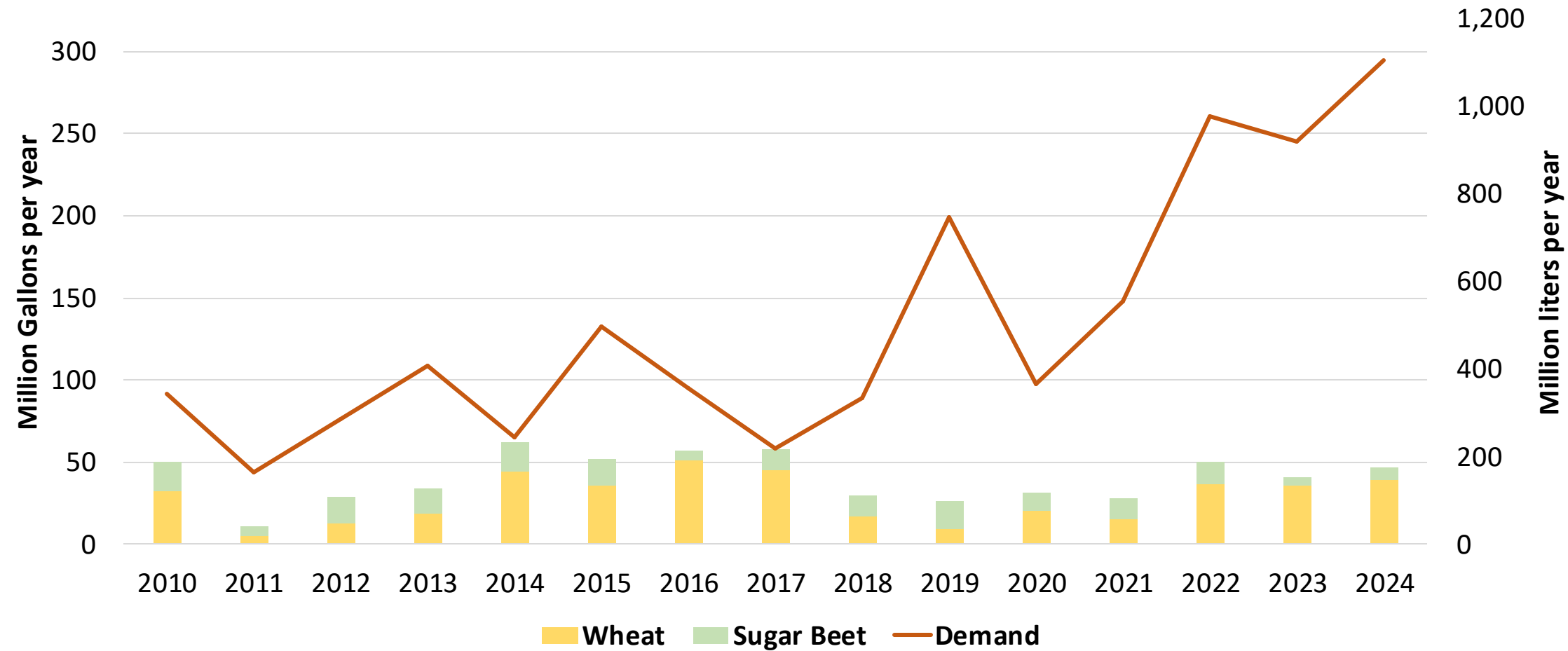
The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexadiene derivative. The structure features a six-membered ring with two double bonds. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.



Ethanol Balance in United Kingdom



Ethanol Demand in the United Kingdom



Source: Eurostat

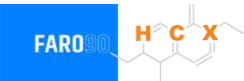
Gasoline Quality in United Kingdom

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commission

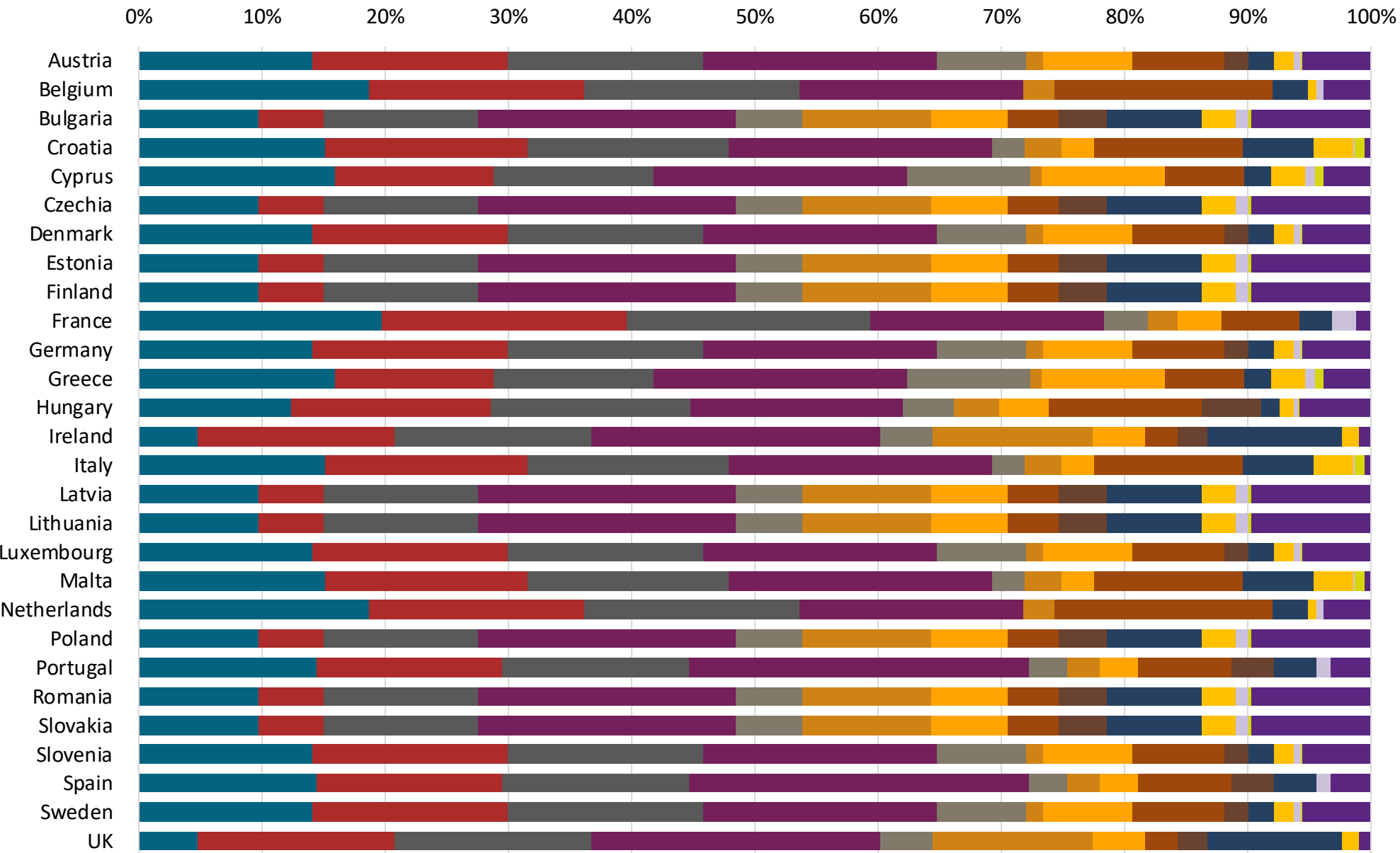
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



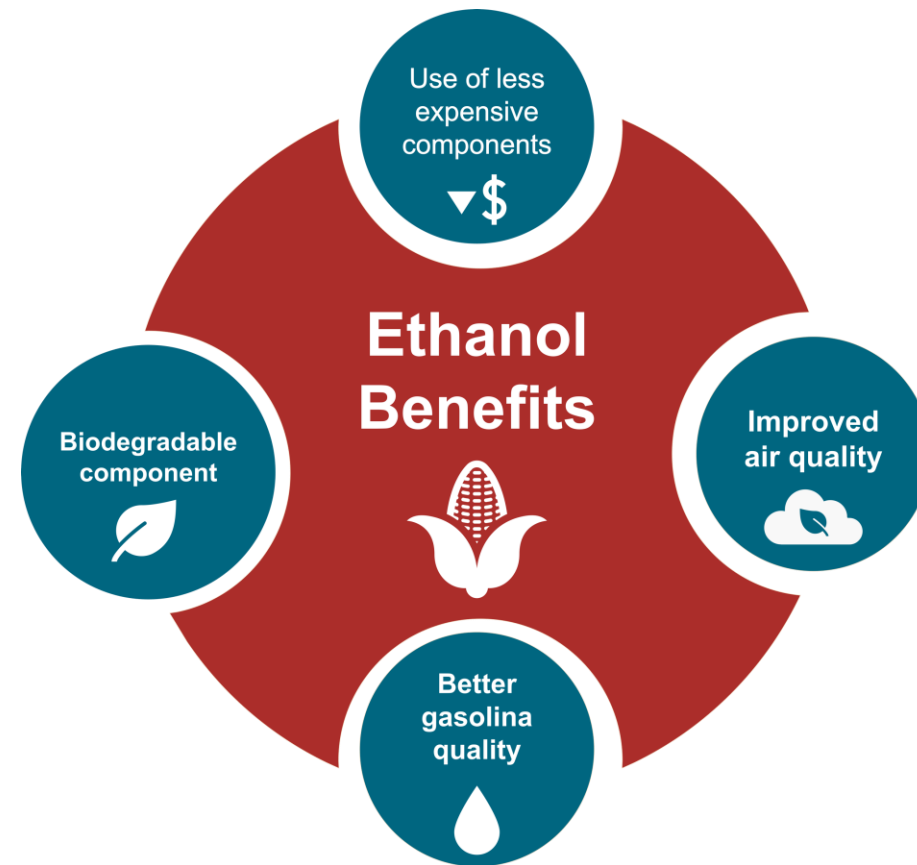
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

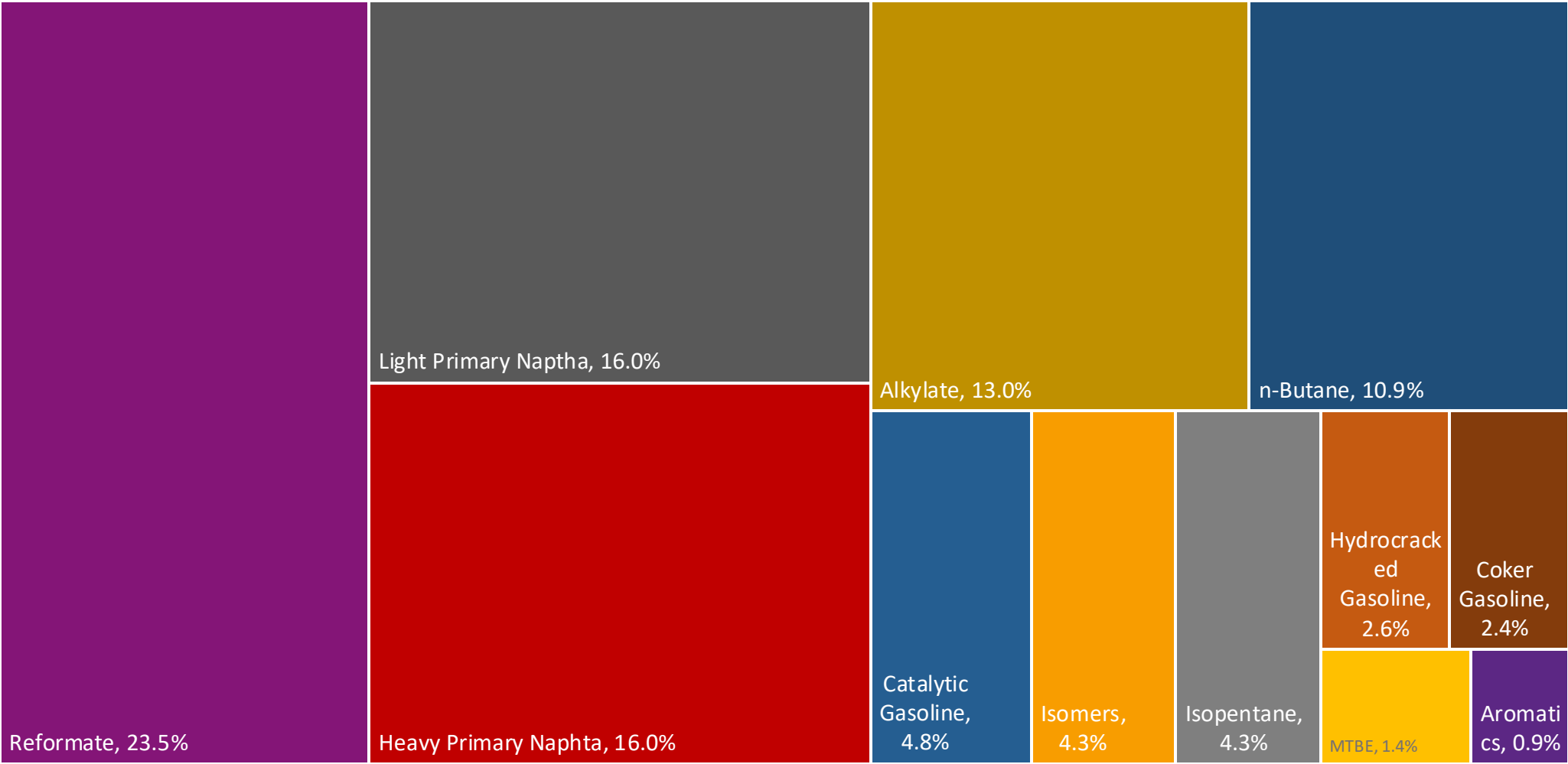
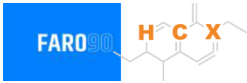
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

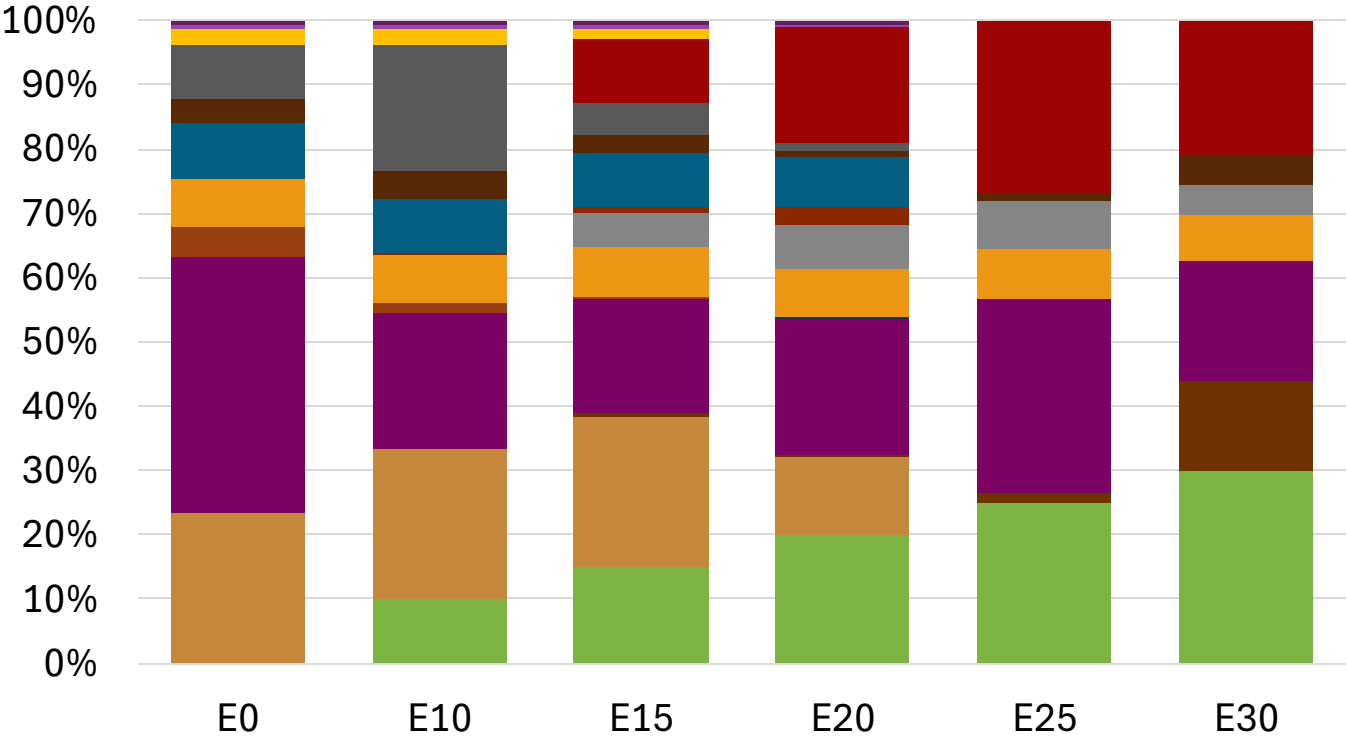
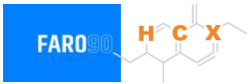
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



United Kingdom Available Blending Components



United Kingdom – Gasoline 95 – Constant Octane

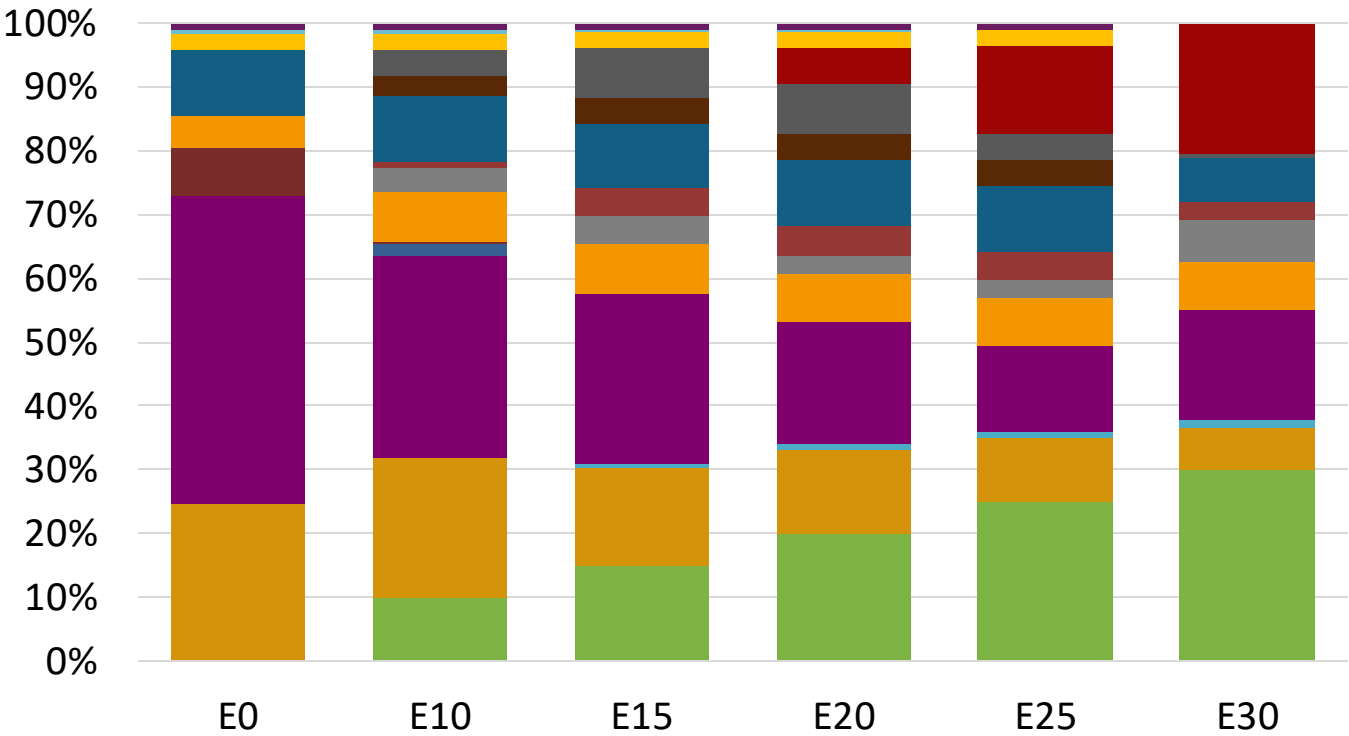
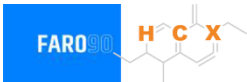


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US\$/gal)	\$ 2.317	\$ 2.187	\$ 2.105	\$ 1.996	\$ 1.886	\$ 1.821

Source: Faro90

United Kingdom - Gasoline 98 - Constant Octane

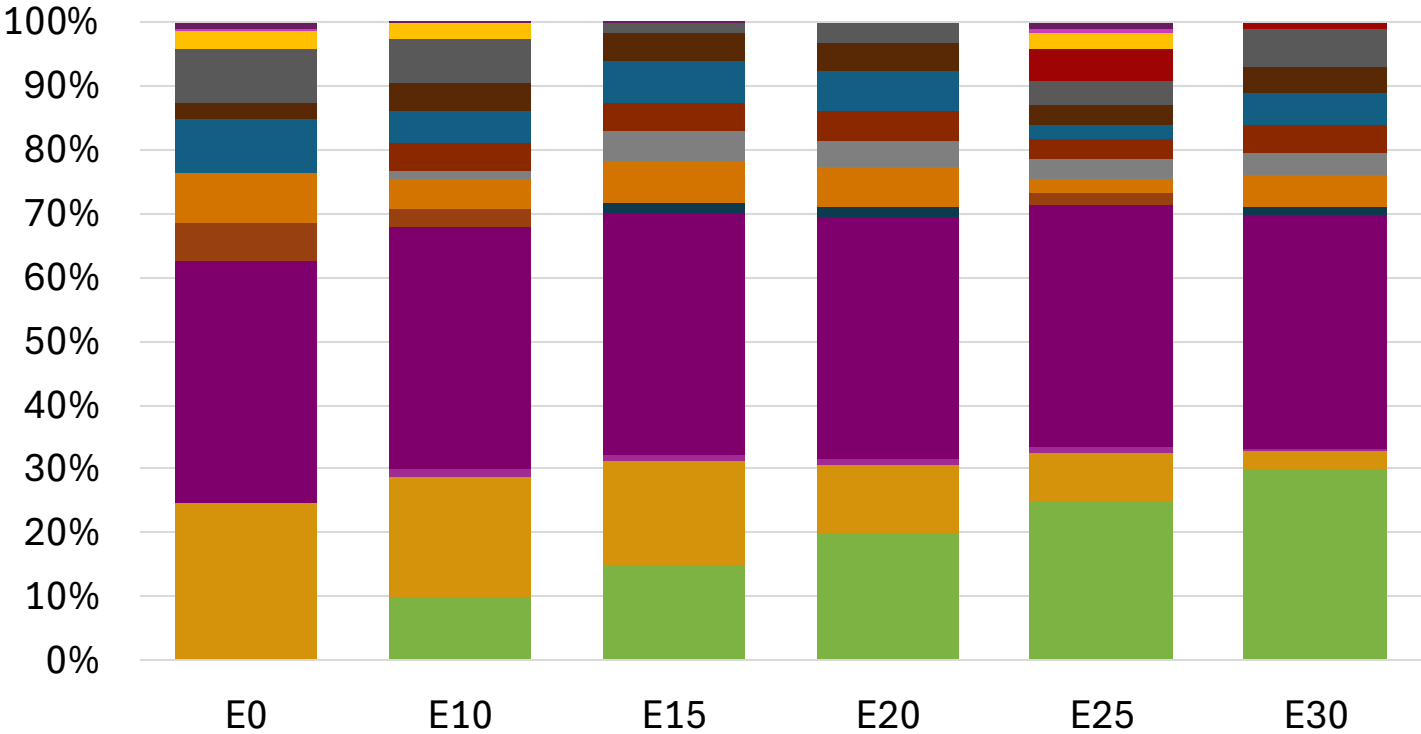
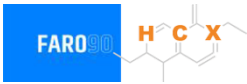


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.440	\$ 2.317	\$ 2.240	\$ 2.157	\$ 2.062	\$ 1.967

Source: Faro90

United Kingdom – Gasoline 95 – Increased Octane

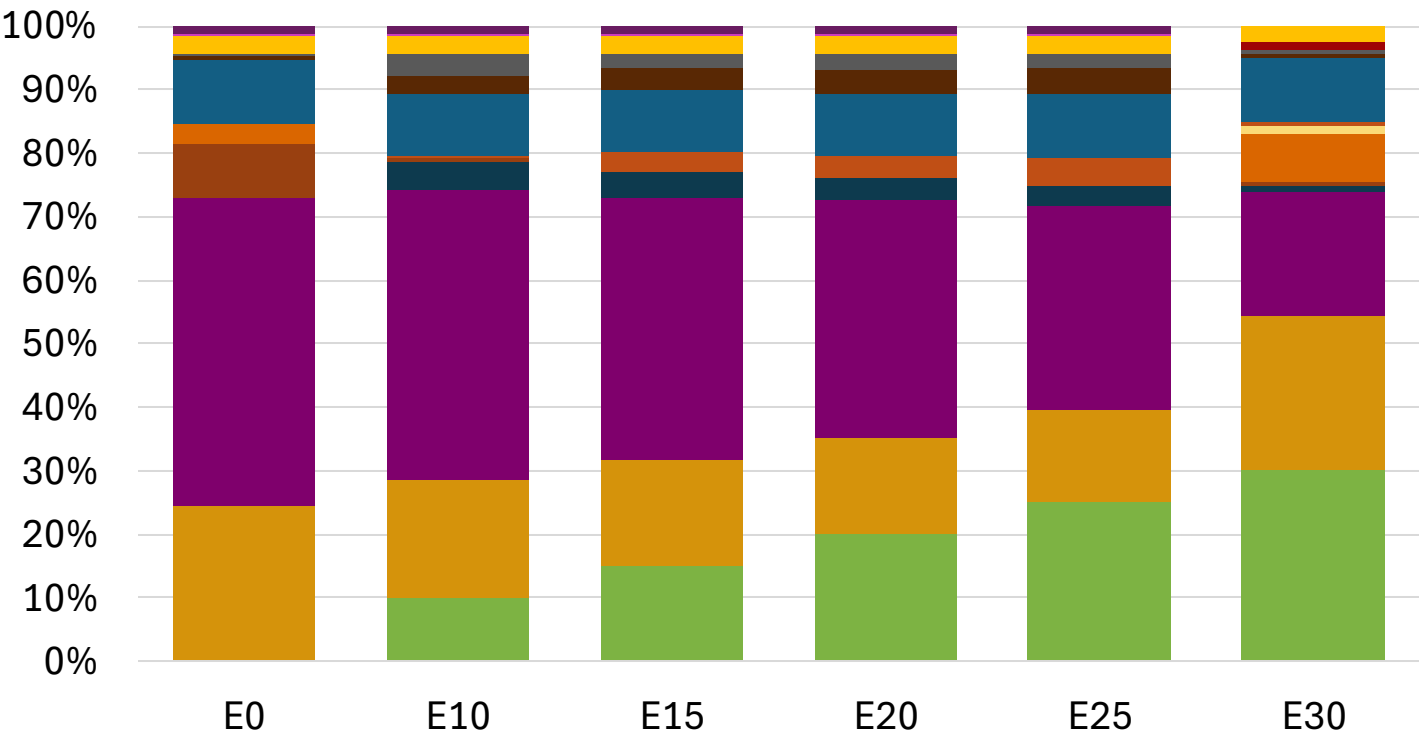
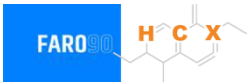


Average CIF 2024
Prices

Octane (RON)	95.0	97.9	99.3	100.9	102.6	103.9
Price (US\$/gal)	\$ 2.301	\$ 2.301	\$ 2.301	\$ 2.301	\$ 2.301	\$ 2.301

Source: Faro90

United Kingdom – Gasoline 98 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	98.0	100.7	102.0	103.3	104.5	105.0
Price (US\$/gal)	\$ 2.425	\$ 2.425	\$ 2.425	\$ 2.425	\$ 2.425	\$ 2.425

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

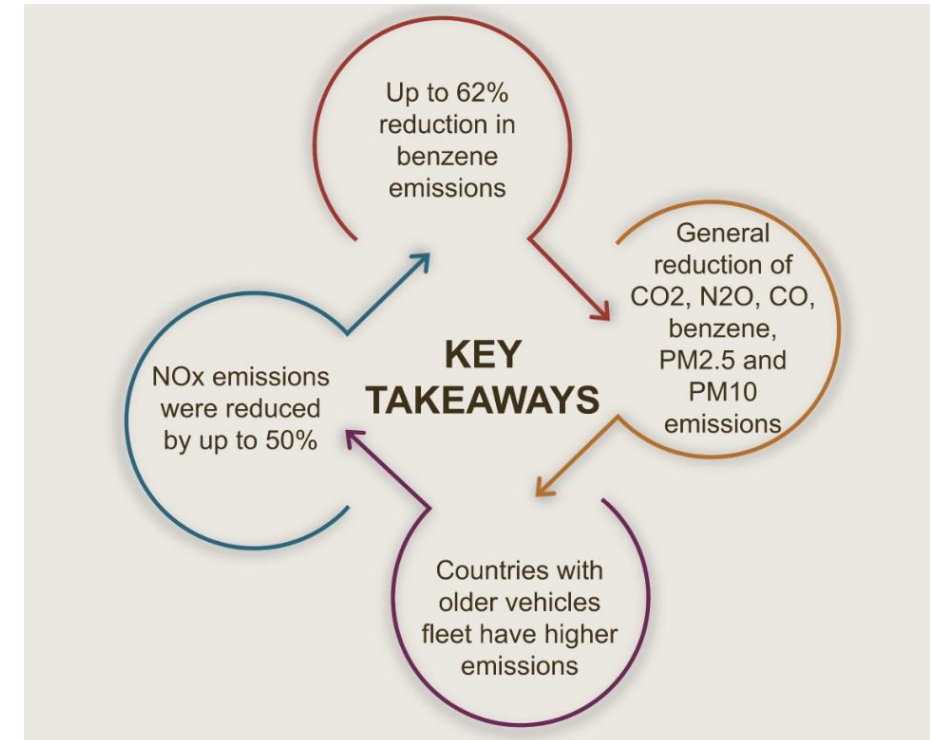
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

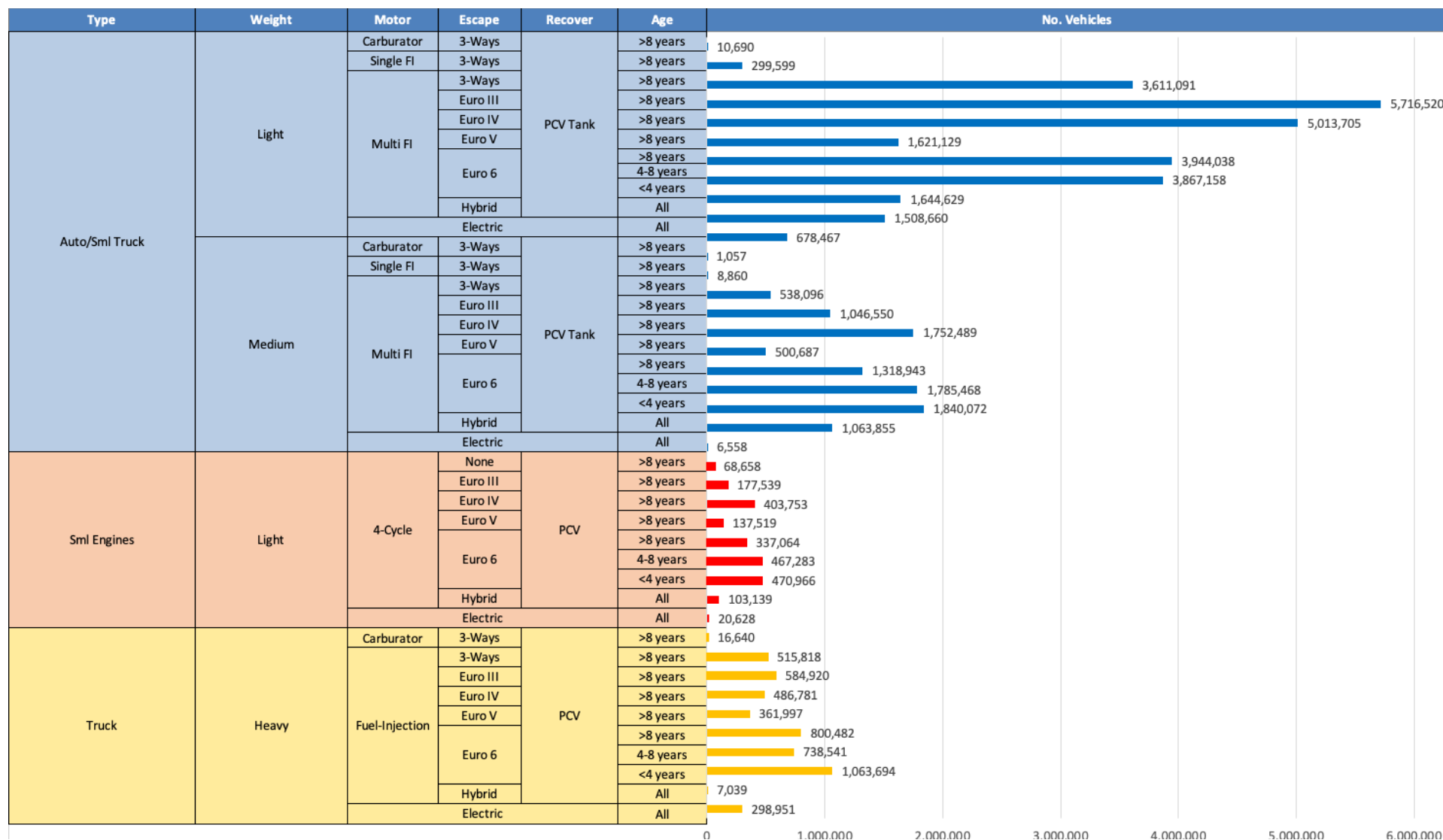
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - United Kingdom

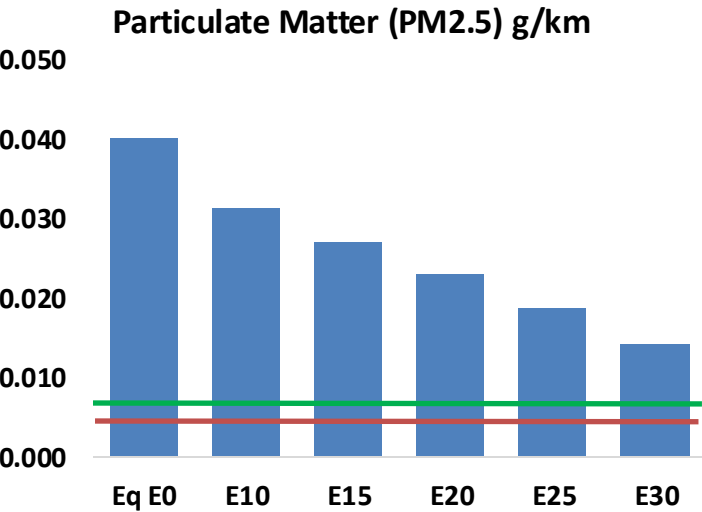
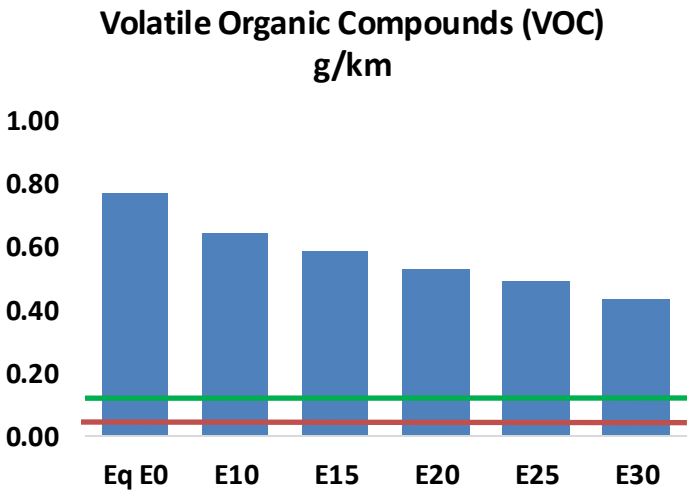
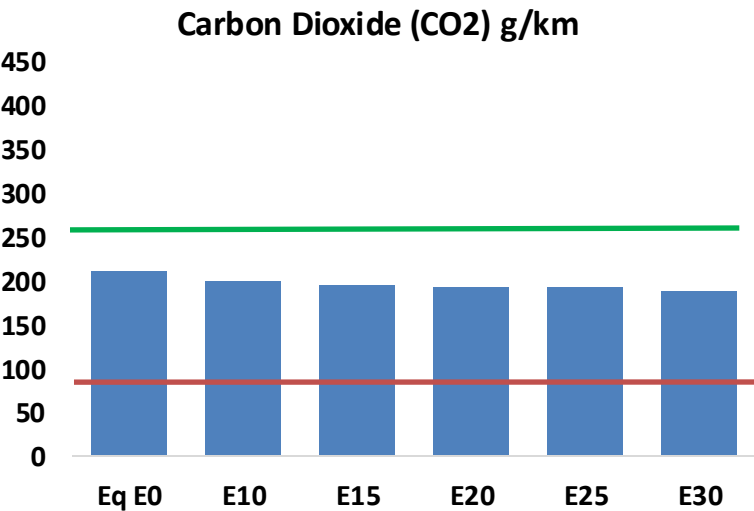
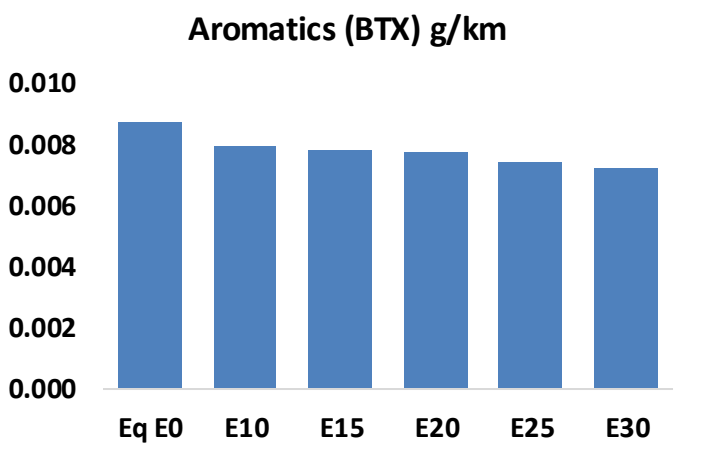
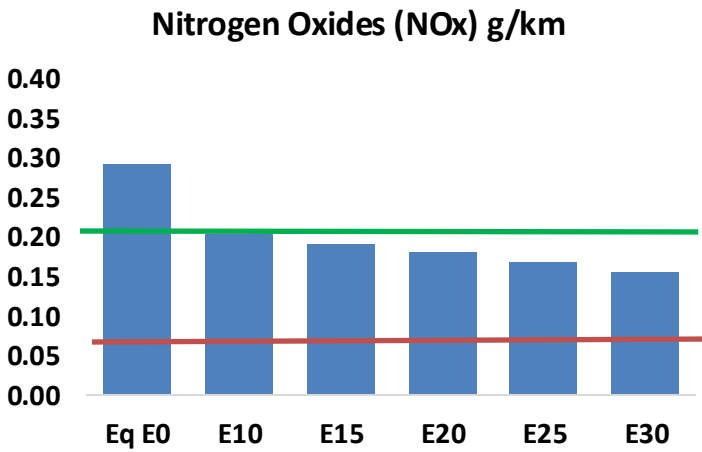
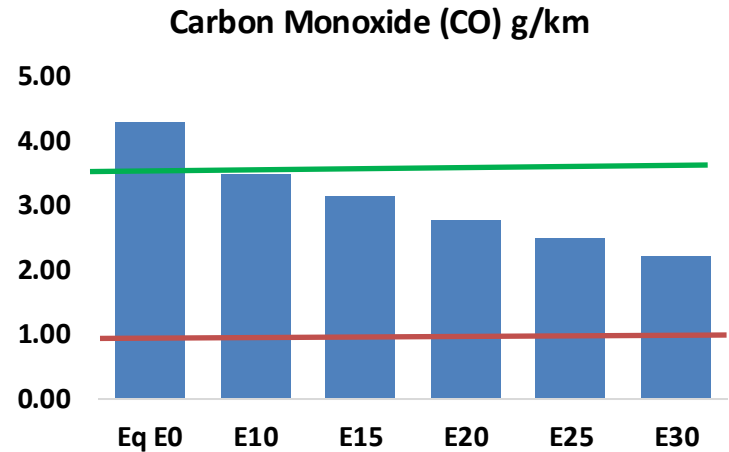
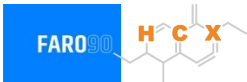


Vehicle Fleet: 44,839,734
Gasoline Vehicle Fleet: 22,834,680

Source: Eurostat, ACEA

United Kingdom – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



United Kingdom – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,592,464	2,347,600	2,102,737	1,893,191	1,678,226	1,511,697	1,337,359	-19%	-35%
CO2	126,780,565	123,520,493	120,441,537	118,020,028	116,821,771	116,113,706	113,973,396	-5%	-8%
VOC	464,487	425,388	386,288	353,587	320,373	294,742	260,863	-17%	-31%
NOx	176,401	132,301	123,481	116,381	109,757	102,454	94,728	-30%	-38%
SOx	1,449	1,339	1,230	1,137	1,048	952	850	-15%	-28%
NH3	46,556	46,095	45,634	45,851	46,366	46,729	47,031	-2%	0%
N2O	2,090	2,079	2,070	2,080	2,123	2,147	2,171	-1%	2%
CH4	93,659	93,659	93,659	95,064	97,125	98,904	100,590	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,440	2,325	2,211	2,131	2,058	2,003	1,924	-9%	-16%
Acetaldehyde	7,685	8,820	12,942	19,708	26,851	30,682	36,254	68%	249%
Formaldehyde	28,374	27,586	32,157	37,760	39,559	43,763	47,641	13%	39%
Benzene	5,291	5,084	4,818	4,741	4,702	4,497	4,352	-9%	-11%
PM2.5	7,836	6,958	6,080	5,280	4,483	3,638	2,757	-22%	-43%
PM10	16,450	14,606	12,763	11,083	9,411	7,637	5,787	-22%	-43%

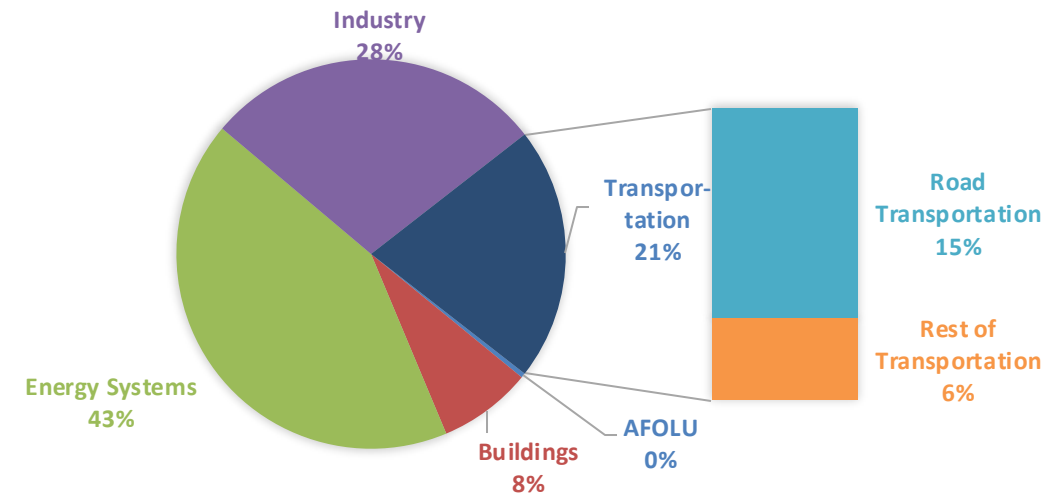
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

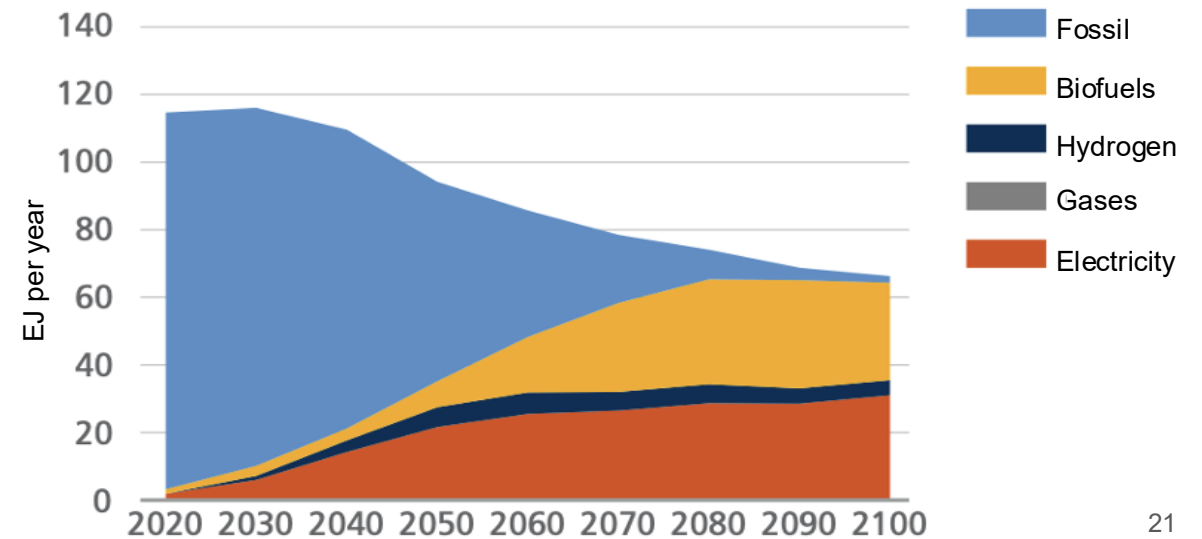
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

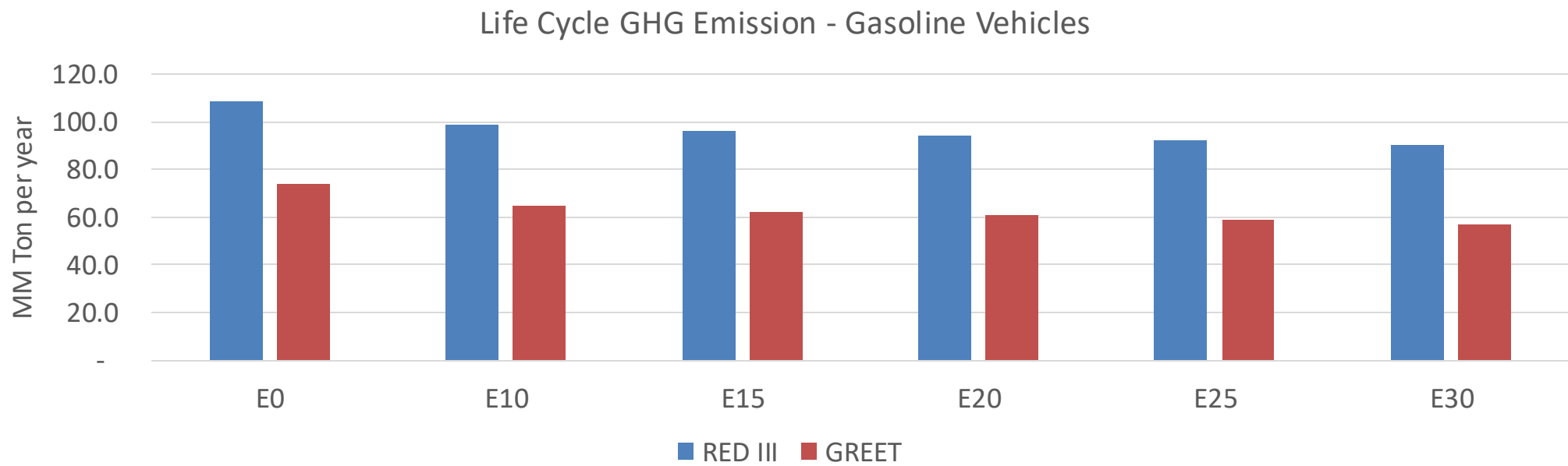
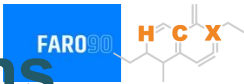
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



United Kingdom – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	402.0
Target @ 2035 (MMT/year)	264.3
Delta	137.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.5%	16.5%	18.0%	20.3%	23.4%
RED II/III	0.0%	9.0%	11.3%	13.1%	14.9%	17.1%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	6.7%	8.9%	9.7%	10.9%	12.6%
RED II/III	0.0%	7.1%	8.9%	10.4%	11.7%	13.5%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90